Lab 9 Reflective Journal

TheBetaThetas | ITAI 1371 | Lab 9: Ensemble Methods

Brandy Gonzales

Before this lab, I didn't know you could basically program a double check with data. In a Random Forest, the system builds many trees, each tree makes its own guess, and the majority vote becomes the final answer.

The most challenging part was when both models scored 100%. I had to look further and inspect why. I learned that both models got 100% because the iris dataset is simple, so it's easy to tell the flowers apart by the length and width of their petals, which made sense once I looked closer.

Real data won't always be this easy, and results won't always be 100%. That's why it's better to use a committee of models. When one model makes a mistake, the majority cancels it out. For our stroke final, combining our top 3 models with voting should give us more dependable predictions than any single model.

Marlon Roberts

In this lab, I learned about the ensemble method, focusing on the difference between a single predictive model and a collective group of models working together to classify the classic Iris flower dataset. Combining multiple individual machine learning models into a "committee" creates a final prediction that is significantly more accurate and less prone to mistakes than relying on one single model. There are 2 ways to build ensemble models, bagging and boosting the models. I trained a RandomForestClassifier and saw how it mitigates the main weakness of a single DecisionTreeClassifier (overfitting) by building 100 diverse trees (n_estimators=100) that vote on the final answer. I also extracted feature_importances_ to reveal exactly which physical measurements of a flower matter most when trying to identify its species. A "committee" of models is better than a single expert model because individual models are highly prone to variance, bias, or overfitting to specific patterns. By averaging their outputs or letting them vote, the random errors of individual models cancel each other out, leading to much better generalization on the unseen data.

Derrick K. Gibbs-McGlaston

This lab helped me understand why combining several models can sometimes make a prediction more reliable. I started with a single Decision Tree and then used a Random Forest with 100 trees. The main difference is that the single tree makes decisions on its own, while the Random Forest combines the votes from many trees. That made the idea of an ensemble easier to understand.

I used the Iris dataset and divided it into training and testing data. Both models used the same split, which made their test results easier to compare. The Decision Tree scored 100.00% and the Random Forest scored 100.00%. These results reminded me that adding more models does not automatically mean a higher score on every dataset. I also learned that accuracy on one test split is only one part of evaluating a model.

The feature importance chart showed that petal width (cm) and petal length (cm) were the most useful measurements for this Random Forest. The chart gave me another way to understand the model beyond looking at accuracy alone. It helped me see that different input features do not always contribute equally to a prediction.

One question I have is whether the results would change if I used a different training and testing split. The workflow connected to earlier labs because I still had to prepare the data, train the model, make predictions, and evaluate the result. The new part was learning how multiple trees work together.

Victor Djuma

In this lab, I learned about ensemble methods and how combining several machine learning models can improve the stability and reliability of predictions. Before this lab, I understood that a model could make predictions by learning patterns from data, but I did not fully understand why using several models together could sometimes be better than relying on only one model. The idea of the “wisdom of the crowd” helped me understand this concept more clearly. If one model makes a mistake, another model may make a different mistake, and when the results are combined, the overall prediction can become stronger.

The first model I worked with was a Decision Tree. I learned that a Decision Tree makes decisions by splitting the data based on different features. It is easy to understand because it follows a step-by-step decision process. However, I also learned that a single Decision Tree can easily overfit the training data. It can learn the training examples too closely instead of learning a general pattern that works well with new data.

After that, I trained a Random Forest model. A Random Forest is an ensemble because it combines many Decision Trees instead of depending on only one tree. Each tree is trained using different samples and features, and the final prediction is based on the combined decisions of all the trees. This helped me understand why Random Forest is generally more stable than a single Decision Tree.

In this lab, both the Decision Tree and the Random Forest achieved 100% accuracy on the test data. At first, I expected the Random Forest to clearly perform better than the single tree because it combines many trees. However, the result showed me that a more advanced model will not always produce a higher accuracy score on every dataset. The Iris dataset is relatively small and the classes are easier to separate, so both models were able to classify the test data correctly. This reminded me that model performance depends on the dataset and not only on how complex the

algorithm is.

Another important part of the lab was feature importance. The Random Forest showed that petal width and petal length were the most important features for classifying the Iris flowers. I found this useful because machine learning is not only about getting a prediction. It is also important to understand which features influence the model’s decision. This can help us better understand the data and explain the model’s results.

Overall, this lab helped me understand the difference between using a single model and using an ensemble of models. I now understand that bagging methods such as Random Forest can improve stability by combining several models. I also learned that accuracy alone does not tell the whole story because two different models can achieve the same accuracy even though they work differently. The biggest lesson I learned from this lab is that ensemble methods use the strengths of multiple models to create a more reliable prediction, while feature importance can help us understand which parts of the data matter most.

Pablo Lara

Not yet received.